Symmetries in Physics — Exercise Sheet 3

Representations and complete reducibility

Variant: Questions only

Exercise 1 Write the three matrices of the regular representation of $\mathbb{Z}/3\mathbb{Z}$ on $\mathbb{C}^3$ and decompose it explicitly into one-dimensional irreducibles, exhibiting the change of basis.

Exercise 2 On $\mathbb{C}^3$, S_3 acts by permuting coordinates. Show that $V = \{x : \, x_1 + x_2 + x_3 = 0\}$ is invariant, that the constants form an invariant complement, and prove directly — no characters — that the two-dimensional representation on V is irreducible.

Exercise 3 Let $\mathbb{Z}/2\mathbb{Z}$ act on $\mathbb{C}^2$ by $g = \left(\begin{smallmatrix} 1 & 1 \\ 0 & -1 \end{smallmatrix}\right)$ (check $g^2 = \mathbb{1}$). Average the standard inner product over the group, verify the result is invariant, and find a basis in which g is unitary.

Exercise 4 Maschke needs finiteness: let the infinite group $\mathbb{Z}$ act on $\mathbb{C}^2$ by $n \mapsto \left(\begin{smallmatrix} 1 & n \\ 0 & 1 \end{smallmatrix}\right)$. Show this is a representation, that the line $\mathbb{C}e_1$ is invariant, and that it has *no* invariant complement. Where exactly does the averaging proof break?

Exercise 5 Let $\mathbb{Z}/2\mathbb{Z}$ act on wave functions $\psi \in L^2(\mathbb{R})$ by the parity operator $(\Pi\psi)(x) = \psi(-x)$. Show directly that the $+1$ and -1 eigenspaces of Π are invariant and are precisely the even and odd functions. Prove that every ψ decomposes uniquely into one function of each kind, and state how the non-identity element acts on the two invariant pieces.